Faculty of Media Engineering and Technology
German University in Cairo

Comparing PageRank and HITS using Apache Spark

A thesis submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering

By

Mark Maged Samir

Supervised by

Islam El Maddah

June 6, 2026

This is to certify that:

(i) the thesis comprises only my original work toward the Bachelor Degree of Science (B.Sc.) at the German University in Cairo (GUC),

(ii) due acknowledgment has been made in the text to all other material used

Mark Maged Samir
June 6, 2026

Acknowledgments

I would like to thank the following for their efforts with me in my study and for tolerating me during this journey...

List of Publications

The following publications have been published while in conduct of this study:

- Authors. paper title. In *conference/journal* Status(under review, accepted to be published,Volume or pages in case of published), Year.

Abstract

Link analysis algorithms constitute a foundational pillar of modern information retrieval, network science, and web intelligence. This thesis presents a rigorous comparative survey of two seminal link analysis algorithms—PageRank and Hyperlink-Induced Topic Search (HITS)—within the context of distributed graph processing using Apache Spark. We provide a comprehensive treatment of the theoretical foundations, mathematical formulations, convergence properties, and inherent limitations of both algorithms. Beyond the classical web search domain, we extend the application of these algorithms to the analysis of real-world complex networks, specifically professional football club transfer networks, wherein clubs are modeled as nodes and player transfers as weighted directed edges. To operationalize this comparative study, we design and implement a full-stack web application, the *Transfer Analyzer*, which leverages a PySpark backend to execute both PageRank and HITS on real transfer market data and exposes interactive visualizations—including ranked bar charts, force-directed network graphs, and chord-diagram transfer flow maps—through a browser-based frontend. The configurable parameters of the application (iterations, damping factor, league and season filters, weighting mode, and analysis target) enable systematic exploration of algorithmic behavior under varying conditions. Our comparative analysis reveals that PageRank, as a query-independent, global ranking mechanism, is better suited for identifying systemically central nodes in large-scale networks, while HITS, with its dual hub–authority decomposition, provides richer structural insight into the directional roles nodes play within localized subgraphs. This thesis contributes both a unified theoretical framework for understanding these algorithms and a practical, reproducible implementation artifact that bridges the gap between abstract graph theory and applied distributed computing.

Keywords: PageRank, HITS, Link Analysis, Apache Spark, Distributed Computing, Graph Algorithms, Football Transfer Networks, Complex Networks, PySpark, Web Application

Contents

List of Abbreviations

List of Figures

List of Tables

Chapter 1

Introduction

The World Wide Web, since its inception, has grown into one of the largest and most structurally complex information systems ever constructed by humanity. By the late 1990s, the exponential proliferation of web pages rendered purely content-based retrieval methods inadequate for distinguishing authoritative sources from the vast sea of marginally relevant or deliberately misleading pages [1]. This fundamental inadequacy motivated a paradigm shift in information retrieval: the exploitation of the hyperlink structure of the web as a proxy for human editorial judgment. A hyperlink from page A to page B was reconceptualized not merely as a navigational convenience but as an implicit endorsement—a vote of confidence in the quality, relevance, or authority of the destination page [2].

This insight gave rise to the field of *link analysis*, which applies graph-theoretic and linear-algebraic techniques to extract meaningful rankings, influence measures, and structural roles from the topology of interconnected networks. Two algorithms, in particular, emerged as the foundational contributions to this field: the **PageRank** algorithm, introduced by Brin and Page in 1998 [1], and the **Hyperlink-Induced Topic Search (HITS)** algorithm, proposed by Kleinberg in 1999 [3]. PageRank assigns a single, global importance score to every node in a directed graph by modeling a random walk with teleportation, while HITS decomposes the roles of nodes into two complementary dimensions—hubs and authorities—computed over a query-specific subgraph.

Although both algorithms were originally conceived for web search ranking, their mathematical generality has enabled their application to a wide array of complex networks beyond the web, including social networks, citation graphs, biological interaction networks, economic systems, and, as explored in this thesis, professional sports transfer markets [4, 5, 6]. Simultaneously, the ever-increasing scale of real-world networks has necessitated the adoption of distributed computing frameworks capable of processing graphs with millions or billions of edges. Apache Spark, with its in-memory computation model, resilient distributed datasets

(RDDs), and specialized graph processing library GraphX, has emerged as a leading platform for large-scale graph analytics [7].

1.1 Motivation

In 1998, Sergey Brin and Larry Page introduced the PageRank algorithm, fundamentally shifting web search from purely textual matching to structural link analysis. By conceptualizing the World Wide Web as a massive directed graph, PageRank operated on the premise that a hyperlink from one page to another acts as an endorsement. This allowed search engines to rank pages based on their objective, topological authority rather than easily manipulated keyword frequencies, significantly improving the quality of information retrieval [1].

Shortly thereafter, in 1999, Jon Kleinberg proposed the HITS algorithm, which took a fundamentally different, query-dependent approach by recognizing that web pages serve two distinct semantic functions: providing authoritative information and providing curated directories of links [3]. The concurrent emergence of these two complementary paradigms established a rich design space for link analysis that continues to inform modern information retrieval, recommendation systems, and network science.

The football transfer market, in which clubs exchange players for monetary fees, naturally forms a weighted directed graph whose analysis via link analysis algorithms can reveal the systemic centrality, market competitiveness, and structural roles of participating clubs [4, 5]. The convergence of algorithmic theory, real-world network applications, and scalable distributed infrastructure forms the tripartite motivation for this thesis.

As articulated in the project brief by Dr. El-Maddah [8], the implementation of PageRank and HITS using Apache Spark GraphX provides students with hands-on experience in distributed systems and large-scale algorithm execution, with applications spanning large-scale recommendation engines, enterprise search tools, fraud detection networks, and large social network analysis. The skills required encompass Apache Spark, GraphX, Scala/Python, distributed computation, and cloud processing [8].

1.2 Problem Statement

Despite the extensive individual study of PageRank and HITS in the literature, a significant gap persists in the availability of unified, interactive, and reproducible comparative frameworks that allow researchers and practitioners to:

(a) execute both algorithms on the same real-world dataset under identical conditions,

(b) systematically vary algorithmic parameters (e.g., damping factor, iteration count, edge weighting scheme) and observe the impact on output rankings, and

(c) visualize the results through multiple complementary modalities—ranked lists, network topologies, and inter-cluster flow diagrams—within a single cohesive tool.

Furthermore, the vast majority of existing comparative studies confine their analysis to web-graph or synthetic datasets [9, 10, 11], leaving the behavior of these algorithms on non-web complex networks—particularly weighted, financially driven networks such as the football transfer market—insufficiently explored. The question of how the global, query-independent ranking of PageRank and the local, dual-score decomposition of HITS differentially characterize the structural roles of nodes in such networks remains open.

This thesis addresses the following research questions:

- **RQ1:** What are the theoretical and computational distinctions between the PageRank and HITS algorithms in terms of their mathematical formulations, convergence guarantees, scalability, and vulnerability to manipulation?
- **RQ2:** How can both algorithms be implemented within a unified distributed computing pipeline using Apache Spark (PySpark) and applied to a real-world weighted directed graph derived from professional football transfer data?
- **RQ3:** What structural insights do PageRank and HITS, respectively, reveal about the football transfer network, and how do these insights differ across configurable parameters such as damping factor, iteration count, and edge weighting mode?

1.3 Objectives

To address the research questions stated above, this thesis pursues the following objectives:

1. **Theoretical Synthesis:** Provide a comprehensive and self-contained exposition of the mathematical formulations of both PageRank and HITS, including their Markov chain and eigenvector interpretations, convergence properties, computational complexity, known limitations, and principal modified variants.
2. **Comparative Analysis:** Develop a rigorous, multi-dimensional comparative framework that contrasts PageRank and HITS along the axes of execution model (offline vs. online), graph scope (global vs. local), score semantics (single score vs. dual hub–authority scores), vulnerability to manipulation, and suitability for different network analysis tasks.
3. **Distributed Implementation:** Design and implement a distributed graph processing pipeline using Apache Spark (PySpark) that ingests real-world football transfer data, constructs a weighted directed graph, and executes both PageRank and HITS with configurable parameters.
4. **Application Artifact:** Develop a full-stack web application—the *Transfer Analyzer*—

that serves as the user-facing interface for the distributed pipeline, exposing interactive visualizations (bar charts, network graphs, transfer flow chord diagrams) and enabling systematic exploration of algorithmic behavior.

5. **Domain Application:** Demonstrate the applicability of link analysis algorithms to non-web complex networks by analyzing the football transfer market and interpreting the resulting scores in terms of club centrality, market influence, and structural role.

1.4 Thesis Outline

The remainder of this thesis is organized as follows:

Chapter 2: Background. This chapter provides the theoretical foundations necessary for the subsequent analysis. Section 2.1 presents the core concepts, including the mathematical formulations of both PageRank and HITS, their modified variants, and the architecture of Apache Spark for distributed graph processing. Section 2.2 surveys the relevant literature across five thematic areas: the evolution of PageRank, the development of HITS, comparative studies of the two algorithms, distributed implementations of link analysis, and graph analytics applied to non-web domains. Section 2.3 describes the football transfer dataset used as the real-world graph in our implementation. Section 2.4 defines the evaluation metrics employed for comparing algorithmic outputs. Section 2.5 synthesizes the reviewed literature and identifies the gap addressed by this thesis.

Chapter 3: Methodology. This chapter details the design and implementation of the comparative study. Section 3.1 provides the formal problem formulation, casting the football transfer market as a weighted directed graph. Section 3.2 describes the distributed graph processing pipeline, from data ingestion through algorithm execution to result aggregation. Section 3.3 presents the experimental design for the comparative analysis. Section 3.4 details the implementation. Section 3.5 presents the architecture and functionality of the Transfer Analyzer web application. Subsequent sections cover the PySpark backend, frontend interface, configurable parameters, algorithm integration, and expected analytical outputs.

Chapter 4: Results. This chapter presents the experimental setting, dataset description, tools used, evaluation metrics, configurable parameters, comparative analysis findings, and the development and testing environment.

Chapter 5: Conclusion and Future Work. This chapter summarizes the contributions of the thesis, discusses the implications of the comparative findings, and outlines directions for future research.

Chapter 2

Background

This chapter establishes the theoretical foundations and contextual literature underpinning the comparative study of PageRank and HITS. We begin with a self-contained presentation of the core concepts (Section 2.1), followed by a structured literature review (Section 2.2), a description of the dataset (Section 2.3), a definition of evaluation metrics (Section 2.4), and a synthesizing discussion (Section 2.5).

2.1 Concepts Overview

2.1.1 Link Analysis and Web Graph Theory

The World Wide Web can be formally represented as a directed graph $G = (V, E)$, where the set of vertices $V = \{v_1, v_2, \ldots, v_N\}$ corresponds to individual web pages and the set of directed edges $E \subseteq V \times V$ corresponds to hyperlinks. An edge $(u, v) \in E$ indicates that page u contains a hyperlink pointing to page v. The *in-degree* of a node v, denoted $\deg^-(v) = |B(v)|$, is the number of edges directed toward v, while the *out-degree*, denoted $\deg^+(v) = |F(v)|$, is the number of edges originating from v. Here, $B(v) = \{u \in V : (u, v) \in E\}$ denotes the set of *predecessors* (or in-neighbors) and $F(v) = \{w \in V : (v, w) \in E\}$ denotes the set of *successors* (or out-neighbors) of v [1, 2].

Link analysis is the discipline concerned with extracting meaningful structural information—rankings, centrality measures, community structures, and influence flows—from the topology of such directed graphs. The central premise is that the pattern of hyperlinks encodes collective human judgment about the relative importance and relevance of web pages. Two principal algorithmic paradigms have dominated this field: PageRank and HITS, which differ fundamentally in their scope, execution model, and the type of structural information they extract [3, 9].

2.1.2 The PageRank Algorithm

In 1998, Sergey Brin and Larry Page introduced the PageRank algorithm, fundamentally shifting web search from purely textual matching to structural link analysis. By conceptualizing the World Wide Web as a massive directed graph, PageRank operated on the premise that a hyperlink from one page to another acts as an endorsement. This allowed search engines to rank pages based on their objective, topological authority rather than easily manipulated keyword frequencies, significantly improving the quality of information retrieval [1].

The Random Surfer Model and Markov Chains

Classical PageRank relies on the "random surfer" model, simulating a user randomly clicking hyperlinks. Modeled as a directed graph $G = (V, E)$, a node u confers a fraction of its importance, $PR(u)/L(u)$ (where $L(u)$ is the out-degree of u), to its destination node v. The iterative propagation of rank is defined as:

$$PR_{i+1}(v) = \sum_{u \in B(v)} \frac{PR_i(u)}{L(u)} \tag{2.1}$$

where $B(v)$ is the set of all nodes pointing to v. This formulation is a discrete-time Markov chain, represented by a transition matrix [1, 2].

The transition matrix $\mathbf{M}$ of this Markov chain is defined such that the entry $M_{ij} = 1/L(j)$ if there exists an edge from node j to node i, and $M_{ij} = 0$ otherwise. The PageRank vector $\boldsymbol{\pi}$ is then the stationary distribution of the Markov chain, satisfying $\boldsymbol{\pi} = \mathbf{M}\boldsymbol{\pi}$—that is, it is the principal eigenvector of $\mathbf{M}$ corresponding to eigenvalue 1 [2].

Damping Factor and Eigenvector Computation

To resolve topological anomalies like "spider traps" and "dangling nodes," PageRank introduces a damping factor, d (typically 0.85), representing the probability that the surfer follows existing links. With probability $1-d$, the surfer teleports to a random page in the network. The modified formula is:

$$PR(v) = \frac{1-d}{N} + d \sum_{u \in B(v)} \frac{PR(u)}{L(u)} \tag{2.2}$$

This transforms the matrix into a strictly positive Google Matrix, ensuring convergence to a unique principal eigenvector via the power iteration method [1, 2].

In matrix notation, the Google Matrix $\mathbf{G}$ is defined as:

$$\mathbf{G} = d\,\mathbf{M} + \frac{1-d}{N}\,\mathbf{1}\mathbf{1}^T \tag{2.3}$$

where $\mathbf{1}$ is the N-dimensional column vector of ones. By the Perron–Frobenius theorem, since $\mathbf{G}$ is a column-stochastic, primitive, and irreducible matrix, it possesses a unique stationary distribution that the power iteration method is guaranteed to converge to [2, 12].

Convergence and Complexity

The algorithm's convergence rate is bounded by the damping factor d. The overall time complexity for computation via the power method is $O(k \times (N + E))$, where N represents nodes, E represents edges, and k represents the iterations required, making it highly scalable for large graphs [1, 2]. In practice, convergence to a tolerance of $\epsilon = 10^{-6}$ typically requires $k \approx 50$–100 iterations for web-scale graphs with $d = 0.85$ [2].

Limitations of the Classical Model

Despite its elegance, classical PageRank has notable limitations [13, 14]:

- **Uniform Teleportation:** The standard model teleports users uniformly across the web, ignoring personalized contexts or specific user intents.
- **Equal Link Weighting:** It assumes all outgoing links have equal semantic value, dividing rank blindly by out-degree.
- **Vulnerability to Manipulation:** The algorithm is highly susceptible to artificial "link farms" created by malicious actors to inflate rankings.
- **Bias Against New Nodes:** Older nodes accumulate more links over time, making it difficult for newly created, highly relevant nodes to surface quickly.

2.1.3 The HITS Algorithm

While PageRank evaluates the web as a single, uniform landscape, the Hyperlink-Induced Topic Search (HITS) algorithm, developed by Jon Kleinberg in 1999, takes a fundamentally different, query-dependent approach. Instead of assigning a single rank to a page, HITS recognizes that web pages serve two distinct semantic functions: providing information or providing directions [3].

Consequently, HITS assigns two separate scores to every node in the network:

- **Authority Score** (a): Measures the quality and relevance of the information on a page. A good Authority is a page that is linked to by many good Hubs.
- **Hub Score** (h): Measures the quality of the links provided by a page. A good Hub is a page that links to many good Authorities.

This creates a mutually reinforcing relationship: a node's authority depends on the hubs pointing to it, and its hub value depends on the authorities it points to [3].

Mathematical Formulation of HITS

The HITS algorithm computes these scores iteratively using a pair of update equations. For a given node v in a directed graph $G = (V, E)$:

The Authority Update Rule:

$$a(v) = \sum_{u \in B_v} h(u) \tag{2.4}$$

where:

- $a(v)$: The Authority score of node v.
- B_v: The set of all nodes u that point to node v (in-links).
- $h(u)$: The Hub score of the incoming node u.

The Hub Update Rule:

$$h(v) = \sum_{w \in F_v} a(w) \tag{2.5}$$

where:

- $h(v)$: The Hub score of node v.
- F_v: The set of all nodes w that node v points to (out-links).
- $a(w)$: The Authority score of the destination node w.

Because these scores are continually added together during each iteration, the numbers can quickly grow to infinity. To solve this, HITS requires a **Normalization** step at the end of every iteration. Both the hub and authority scores of all nodes are divided by the sum (or the square root of the sum of squares) of all scores in the network, ensuring the values remain stable and converge [3, 9].

Matrix Interpretation

The HITS update rules can be expressed in matrix form. Let $\mathbf{A}$ be the $N \times N$ adjacency matrix of the directed graph, where $A_{ij} = 1$ if there is an edge from node i to node j. Let $\mathbf{a}$ and $\mathbf{h}$ denote the authority and hub score vectors, respectively. Then the update rules become [3]:

$$\mathbf{a}^{(k+1)} = \mathbf{A}^T \mathbf{h}^{(k)}, \qquad \mathbf{h}^{(k+1)} = \mathbf{A}\, \mathbf{a}^{(k+1)} \tag{2.6}$$

Substituting the first equation into the second yields:

$$\mathbf{a}^{(k+1)} = \mathbf{A}^T \mathbf{A}\, \mathbf{a}^{(k)}, \qquad \mathbf{h}^{(k+1)} = \mathbf{A}\mathbf{A}^T\, \mathbf{h}^{(k)} \tag{2.7}$$

Thus, the authority vector converges to the principal eigenvector of $\mathbf{A}^T\mathbf{A}$, and the hub vector converges to the principal eigenvector of $\mathbf{A}\mathbf{A}^T$ [3]. These are, equivalently, the left and right singular vectors of $\mathbf{A}$ corresponding to the largest singular value, establishing a deep connection between HITS and singular value decomposition (SVD) [9].

Operational Procedure

Unlike PageRank, which is computed over the entire web graph offline, HITS operates on a focused subgraph constructed in response to a specific user query. The operational procedure consists of three phases [3]:

1. **Root Set Construction:** A traditional text-based search engine returns a set of pages relevant to the query. This is the *root set* R.
2. **Base Set Expansion:** The root set is expanded to include all pages that either link to or are linked from any page in R, forming the *base set* $S \supseteq R$.
3. **Iterative Computation:** The hub and authority scores are iteratively computed over the subgraph induced by S until convergence.

2.1.4 Modified PageRank Variants

To address the shortcomings of the classical PageRank model enumerated in Section 2.1.2, researchers have developed advanced variants:

- **Personalized PageRank (PPR):** Replaces the uniform teleportation vector with a custom probability distribution vector tailored to individual user interests, confining rank to a specific topological neighborhood [15]. Formally, the teleportation component $\frac{1-d}{N}$ is replaced by $(1-d) \cdot \mathbf{p}$, where $\mathbf{p}$ is a personalization vector with $\|\mathbf{p}\|_1 = 1$.
- **Topic-Sensitive PageRank:** Uses precomputed rank vectors biased toward representative basis topics. It calculates a query-sensitive score using a linear combination of these topic-specific ranks to improve precision [16]. Given K topic categories, K separate PageRank vectors $\{\boldsymbol{\pi}_1, \ldots, \boldsymbol{\pi}_K\}$ are precomputed, and at query time, the final score is $\boldsymbol{\pi}_q = \sum_{j=1}^{K} c_j \boldsymbol{\pi}_j$, where the coefficients c_j reflect the query's topical distribution.
- **Weighted PageRank (WPR):** Replaces equal link division by allocating rank dynamically based on the relative popularity (in-links and out-links) of the destination nodes [17, 18]. This variant is of particular relevance to the present thesis, as the football transfer network inherently possesses weighted edges corresponding to transfer fees.

- **TrustRank:** Combats link spam by propagating trust outward from a rigorously vetted "seed set" of highly reputable nodes, neutralizing artificially inflated rank structures [19].

2.1.5 Distributed Graph Processing

Real-world graphs—the web graph, social networks, and large-scale transaction networks—routinely contain billions of nodes and edges, far exceeding the memory and computational capacity of a single machine. Distributed graph processing frameworks address this challenge by partitioning the graph across a cluster of machines and executing algorithms in parallel, with inter-machine communication for edge-boundary updates [7, 20].

The two dominant programming models for distributed graph processing are:

1. **Bulk Synchronous Parallel (BSP):** Exemplified by Google's Pregel [20] and Apache Giraph, BSP organizes computation into a sequence of *supersteps*. In each superstep, every vertex executes a user-defined function, sends messages to its neighbors, and synchronizes with all other vertices before the next superstep begins.

2. **MapReduce-Based Models:** Exemplified by Apache Spark's GraphX [7], these models express graph algorithms as sequences of distributed data transformations (map, filter, join, reduce) over vertex and edge tables, leveraging the in-memory caching and fault tolerance of the underlying dataflow engine.

2.1.6 Apache Spark and GraphX

Apache Spark is an open-source, general-purpose distributed computing framework designed for large-scale data processing [7]. Its core abstraction is the **Resilient Distributed Dataset (RDD)**, an immutable, partitioned collection of objects that can be operated on in parallel across a cluster. Spark extends the MapReduce model with in-memory caching, enabling iterative algorithms—such as PageRank and HITS—to avoid the costly disk I/O that hampers Hadoop MapReduce performance [7].

GraphX is Spark's library for graph-parallel computation. It extends the RDD abstraction with a **Graph** object, composed of a vertex RDD and an edge RDD, and provides a set of fundamental graph operators (`mapVertices`, `mapEdges`, `subgraph`, `joinVertices`, `aggregateMessages`) and optimized implementations of common graph algorithms, including PageRank [7].

The **PySpark** API exposes Spark's functionality through Python, enabling the implementation of custom graph algorithms using Spark's distributed DataFrames and RDDs without requiring Scala or Java. For this thesis, PySpark serves as the primary implementation language, enabling seamless integration with a Python-based web application backend.

The key architectural properties of Spark that make it well-suited for iterative graph algorithms include:

- **In-Memory Computation:** Intermediate results from each iteration are cached in memory, eliminating redundant disk reads and dramatically accelerating iterative convergence [7].
- **Lazy Evaluation:** Transformations are recorded as a lineage graph (DAG) and executed only when an action triggers materialization, enabling Spark's optimizer to plan efficient execution strategies.
- **Fault Tolerance:** Lost partitions are automatically recomputed from the lineage DAG, without requiring explicit checkpointing for short computations.
- **Horizontal Scalability:** The framework scales linearly by adding worker nodes to the cluster, enabling the processing of arbitrarily large graphs.

2.1.7 Complex Network Applications

Beyond web search, PageRank is instrumental in modeling complex socioeconomic ecosystems. The mathematical generality of link analysis algorithms—requiring only a directed graph as input—permits their application to any domain in which entities and directed relationships can be identified [4, 5, 6].

Football Player Transfer Networks

The professional football transfer market can be formally modeled as a directed graph $G_{FTN} = (V, E, W)$, where nodes V are clubs, edges E are player transfers, and the weight matrix W represents monetary transfer fees. Evaluating this weighted network via PageRank measures a club's global "coreness" and systemic market competitiveness. High PageRank centrality correlates strongly with a club's ability to attract top-tier talent and its eventual on-pitch performance [4, 5].

To accurately model this financial dynamic, the standard algorithm is adapted into a Weighted PageRank formula:

$$PR(u) = (1 - d) + d \sum_{v \in B_u} \left(PR(v) \times \frac{W(v, u)}{\sum_{w \in F_v} W(v, w)} \right) \tag{2.8}$$

Where the variables are defined as follows:

- $PR(u)$: The new PageRank score of the destination club u (the Buyer).

- d: The Damping Factor. In standard implementations, this is 0.85. It represents the probability that the algorithm continues following links rather than jumping randomly.
- $(1 - d)$: The Teleportation baseline. In standard implementations, this is 0.15. It guarantees every club gets a baseline score of 0.15 per iteration so no club drops entirely to zero.
- B_u: The set of all clubs that link to club u (all clubs that sold a player to club u).
- $PR(v)$: The current PageRank score of the source club v (the Seller).
- $W(v, u)$: The mathematical Weight (Transfer Fee) assigned to the specific edge between club v and club u.
- $\sum_{w \in F_v} W(v, w)$: The sum of all outgoing weights from club v. (The total amount of transfer fees club v received across all its sales).

Employee Mobility Networks

Similarly, corporate ecosystems can be modeled as employee mobility networks $G_{EMN} = (V, E, W)$, where nodes are companies and edges represent employee migration. Edges can be weighted by proxy metrics like seniority gains or salary. In this context, PageRank identifies organizations with maximal global competitiveness. High-PageRank firms benefit disproportionately from systemic knowledge spillovers, successfully attracting strategic human capital and industry trade secrets [6, 21, 22].

2.2 Literature Review

2.2.1 PageRank: Origins and Evolution

The PageRank algorithm was first formally described by Brin and Page in their seminal 1998 paper, "The Anatomy of a Large-Scale Hypertextual Web Search Engine" [1]. The original formulation was subsequently refined in the Stanford InfoLab technical report, which provided a detailed mathematical treatment of the convergence properties of the power iteration method and the spectral properties of the Google Matrix [2]. The algorithm's elegant reduction of the web ranking problem to a well-understood problem in linear algebra—computing the principal eigenvector of a stochastic matrix—was instrumental in establishing Google as the dominant search engine of the early 2000s.

Subsequent research explored the sensitivity of PageRank to the damping factor. Boldi et al. demonstrated that while $d = 0.85$ yields good empirical results for web search, the choice of d fundamentally governs the tradeoff between the influence of the global teleportation component and the local link structure. Higher values of d increase the weight placed on link structure but slow convergence [23].

Langville and Meyer provided a comprehensive linear-algebraic treatment of PageRank, situating it within the broader theory of Markov chains and connecting it to classical results in matrix analysis. Their work clarified the relationship between the spectral gap of the Google Matrix and the convergence rate of the power iteration, establishing that the number of iterations required for convergence scales as $O(\log(N)/\log(1/d))$ [24].

The study of PageRank on networks with special topological properties has also attracted significant attention. Csató and Tóth analyzed the behavior of PageRank on "gluing networks"—graphs formed by connecting smaller component graphs—and derived analytic expressions for the PageRank distribution in terms of the component structures [12].

2.2.2 HITS: Hubs, Authorities, and Query-Dependent Analysis

Kleinberg's 1999 paper, "Authoritative Sources in a Hyperlinked Environment" [3], introduced the HITS algorithm as an alternative to the global ranking paradigm of PageRank. The key conceptual innovation was the recognition that web pages serve distinct structural roles—some pages are primarily repositories of authoritative content, while others function as curated directories (hubs) that aggregate links to authoritative pages.

The HITS algorithm's query-dependent nature represented both its principal strength and its primary limitation. By restricting computation to a focused subgraph relevant to a specific query, HITS could deliver more topically precise results than a global ranking [3]. However, this same restriction made the algorithm vulnerable to *topic drift*—the phenomenon whereby a small number of densely connected but off-topic nodes dominate the subgraph and distort the authority and hub scores [9, 11].

Bharat and Henzinger provided an early empirical analysis of HITS and identified the topic drift problem, proposing mitigations that included weighting the edges of the base set by the textual similarity between the linked pages. Subsequent work by Chakrabarti et al. proposed extensions that combined link analysis with content analysis to improve the topical coherence of HITS results [9].

2.2.3 Comparative Studies of PageRank and HITS

The comparative analysis of PageRank and HITS has been the subject of numerous studies spanning over two decades. The key dimensions of comparison that emerge from the literature are execution model, graph scope, computational cost, vulnerability to manipulation, and output semantics.

Duhan and Sharma provided one of the earliest comprehensive comparative surveys, systematically contrasting the two algorithms on the dimensions of computation time, spam resilience, and query dependence. Their analysis concluded that PageRank's offline, global computation makes it more robust to manipulation and more efficient at query time, while HITS provides richer structural information but at the cost of increased latency and spam vulnerability [9].

A comparative study published in the International Journal of Advanced Research in Computer and Communication Engineering (IJARCCE) [9] evaluated both algorithms on synthetic and real-world web graph datasets, confirming that PageRank demonstrates superior stability across varying graph topologies, while HITS excels in identifying topically coherent clusters of hubs and authorities. The study further noted that the dual hub–authority decomposition of HITS provides information that is fundamentally inaccessible to PageRank, which collapses all structural roles into a single scalar score.

Research published in the International Journal of Engineering Research and Technology (IJERT) [10] extended the comparison to include computational scalability, demonstrating that while both algorithms have comparable asymptotic complexity per iteration—$O(N + E)$—HITS incurs additional overhead due to the base set expansion step and the need for two-phase (hub and authority) updates per iteration. However, the smaller size of the HITS subgraph relative to the full web graph typically more than compensates for this per-iteration overhead.

A more recent comparative analysis published in the International Journal of Computer Applications (IJCA) in 2024 [11] revisited the comparison in the context of modern web architectures, which are characterized by dynamic content, social media integration, and mobile-first design. The study argued that neither algorithm in its classical form is sufficient for contemporary web ranking, but that their underlying principles—global topological authority (PageRank) and local hub–authority decomposition (HITS)—remain foundational building blocks for modern hybrid ranking systems.

2.2.4 Distributed Implementations of Link Analysis

The need to compute PageRank and HITS on web-scale graphs motivated the development of distributed implementations from the earliest days of both algorithms. The original PageRank computation at Google was implemented on a distributed cluster of commodity machines using a custom MapReduce-like framework [1].

The advent of Apache Hadoop's MapReduce provided the first widely accessible platform for distributed PageRank computation. However, the repeated reading and writing of intermediate results to disk between MapReduce jobs made Hadoop poorly suited for the iterative nature of both PageRank and HITS [7].

Apache Spark's introduction of in-memory RDDs represented a transformative improvement for iterative graph algorithms. Gonzalez et al. demonstrated that Spark's GraphX library could compute PageRank on web-scale graphs up to $100\times$ faster than Hadoop-based implementations, primarily due to the elimination of inter-iteration disk I/O. The GraphX programming model expresses graph algorithms as sequences of join and aggregation operations on vertex and edge tables, which the Spark optimizer can parallelize and pipeline efficiently [7].

2.2.5 Graph Analytics in Non-Web Domains

The application of link analysis algorithms to non-web domains has expanded dramatically over the past two decades, reflecting the growing recognition that many real-world systems can be productively modeled as directed graphs.

Football Transfer Networks

Liu et al. were among the first to apply network analysis techniques to the football transfer market, constructing a directed graph of transfers among European clubs and computing various centrality measures, including PageRank [4]. Their analysis revealed a core–periphery structure in the transfer network, with a small number of elite clubs (e.g., Manchester United, Real Madrid, FC Barcelona) forming a densely interconnected core that disproportionately attracts talent from a larger periphery of lower-ranked clubs. Subsequent studies published in PMC [4], IEICE Transactions [5], and ResearchGate [25] corroborated these findings and extended them to include temporal dynamics, demonstrating that the transfer network evolves over time in response to regulatory changes, financial fair play rules, and economic shocks.

Employee Mobility Networks

A parallel line of research has applied graph-theoretic methods to the analysis of employee mobility between firms. Studies published on arXiv [6], EconStor [21], and the Duke Economics Working Paper series [22] have modeled inter-firm employee flows as directed graphs and used PageRank and related centrality measures to identify firms that function as “talent magnets” or “talent exporters” within an industry ecosystem. High-PageRank firms in these networks have been shown to correlate with superior financial performance, innovation output, and market valuation, suggesting that the ability to attract human capital is both a cause and consequence of organizational success.

2.3 Datasets

The primary dataset employed in this thesis is a comprehensive record of professional football player transfers across major European leagues. The dataset encompasses multiple seasons and leagues, including the English Premier League, La Liga (Spain), Serie A (Italy), Bundesliga (Germany), Ligue 1 (France), Liga NOS (Portugal), Eredivisie (Netherlands), the Jupiler Pro

League (Belgium), the English Championship, and several other national and international competitions.

Each record in the dataset corresponds to an individual transfer event and contains the fields described in Table 2.1.

Table 2.1: Dataset fields and their roles in the graph construction.

Field	**Description**	**Role in Graph**
Selling Club	The club transferring the player out	Source node
Buying Club	The club acquiring the player	Destination node
Transfer Fee (€)	The monetary amount of the transfer	Edge weight
Season	The football season of the transfer	Temporal filter
League	The league of the buying/selling club	Categorical filter
Player Name	The name of the transferred player	Edge metadata

The graph is constructed by aggregating individual transfer records into a directed edge list. Each unique club constitutes a node. A directed edge from club v to club u indicates that club v sold at least one player to club u. In the weighted variant, the edge weight $W(v, u)$ is the total monetary value (in euros) of all transfers from v to u within the selected time window.

The dataset supports multiple filtering dimensions—by league, by season range, and by weighting mode (transfer fee in euros or unweighted transfer count)—enabling the user to analyze the graph structure under varying conditions.

2.4 Evaluation Metrics

The comparative evaluation of PageRank and HITS in this thesis employs the following metrics and analytical dimensions:

1. Rank Score Distribution. For PageRank, we analyze the distribution of the scalar PageRank scores $PR(v)$ across all nodes. For HITS, we separately analyze the distributions of authority scores $a(v)$ and hub scores $h(v)$. Visualization via ranked bar charts enables direct identification of the top-N nodes by each score.

2. Rank Correlation. To quantify the agreement or disagreement between the rankings produced by PageRank and HITS (authority), we employ Spearman's rank correlation coefficient ρ, defined as:

$$\rho = 1 - \frac{6\sum_{i=1}^{N} d_i^2}{N(N^2 - 1)} \tag{2.9}$$

where d_i is the difference between the rank of node i under PageRank and its rank under HITS authority. A value of $\rho = 1$ indicates perfect agreement, $\rho = 0$ indicates no correlation, and $\rho = -1$ indicates perfect inversion.

3. Hub–Authority Asymmetry. HITS uniquely provides two scores per node, enabling the identification of nodes that function primarily as hubs (high h, low a) or primarily as authorities (high a, low h). This asymmetry is not captured by PageRank and constitutes a qualitative advantage of HITS for structural role analysis.

4. Convergence Rate. For a fixed tolerance threshold ϵ, the number of iterations required for each algorithm to converge provides a measure of computational efficiency. In the context of our implementation, the user configures a fixed number of iterations (default: 10), and the stability of scores across the final iterations indicates the degree of convergence achieved.

5. Sensitivity to Parameters. The sensitivity of each algorithm's output to configurable parameters—damping factor d for PageRank, iteration count k for both algorithms, and weighting mode (fee-weighted vs. unweighted) for both algorithms—provides insight into the robustness and stability of the rankings.

6. Network Topology Metrics. Complementary graph metrics—including in-degree centrality, out-degree centrality, betweenness centrality, and clustering coefficient—provide context for interpreting the PageRank and HITS scores and assessing the extent to which the algorithmic rankings capture structural properties beyond simple degree.

2.5 Discussion

The literature reviewed in the preceding sections reveals several key themes that inform the design of the comparative study presented in this thesis.

First, **PageRank and HITS embody fundamentally different philosophies of ranking**. PageRank computes a single, global, query-independent importance score that reflects a node's overall centrality in the entire network. HITS computes two query-dependent scores that capture the distinct structural roles of hubs and authorities within a localized subgraph. These philosophies are complementary rather than competing: PageRank answers the question "How globally important is this node?" while HITS answers the question "What role does this node play in this specific context?" [9, 10, 11].

Second, **the mathematical formulations of both algorithms are grounded in spectral graph theory**. PageRank is the principal eigenvector of the Google Matrix; HITS authority and hub vectors are the principal eigenvectors of $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}\mathbf{A}^T$, respectively. This shared spectral foundation enables a unified theoretical treatment while highlighting the structural differences in the matrices involved [2, 3].

Third, **distributed implementation is essential for scalability**. Both algorithms are inherently iterative, requiring repeated global communication (rank propagation) across the graph. Apache Spark's in-memory computation model, combined with GraphX's graph-parallel operators, provides an efficient and scalable platform for both algorithms [7, 8].

Fourth, **non-web domains provide fertile ground for link analysis**. The football transfer network, employee mobility networks, and other socioeconomic graphs exhibit structural properties—such as core–periphery organization, heavy-tailed degree distributions, and community structure—that are well-suited to analysis via PageRank and HITS [4, 5, 6, 21, 22].

Finally, **a significant gap exists in the availability of unified, interactive tools** that enable side-by-side comparison of PageRank and HITS on the same dataset under identical conditions, with configurable parameters and multiple visualization modalities. The Transfer Analyzer application developed in this thesis is designed to fill precisely this gap.

Chapter 3

Methodology

This chapter presents the methodological framework adopted in this thesis for comparing PageRank and HITS on a real-world football transfer network. The methodology combines formal graph modeling, distributed data processing using PySpark, algorithm implementation, experimental design, and the development of an interactive web-based analysis environment. The chapter proceeds from the abstract problem formulation to the concrete implementation of the *Transfer Analyzer* system and the experimental protocol used to interpret the obtained rankings.

3.1 Problem Formulation

The central objective of this thesis is to compare the behavior of PageRank and HITS when applied to the same weighted directed network under controlled and configurable conditions. The selected application domain is the professional football transfer market, in which clubs exchange players across leagues and seasons for monetary transfer fees.

Definition 3.1 (Football Transfer Network)**.** *The football transfer network is modeled as a weighted directed graph*

$$G = (V, E, W),$$

where:

- V *is the set of football clubs;*
- $E \subseteq V \times V$ *is the set of directed edges such that an edge* (u, v) *exists if club* u *transferred at least one player to club* v*;*
- $W : E \to \mathbb{R}^+$ *is a weight function assigning a non-negative value to each edge.*

In the context of this thesis, the direction of each edge is defined from the *selling club* to the *buying club*. This choice preserves the economic meaning of transfer flow: talent and market value move from source club to destination club.

Two edge-weighting schemes are considered:

1. **Fee-weighted mode:** the weight of an edge is the total transfer fee aggregated across all transfers between the same ordered pair of clubs.
2. **Count-weighted mode:** the weight of an edge is the number of transfer events between the same ordered pair of clubs.

The problem addressed by this thesis may therefore be stated as follows: given a filtered football transfer graph G', compute and compare the rankings produced by PageRank and HITS, analyze how the rankings vary under changes in weighting mode and filtering conditions, and provide interactive visual evidence that supports the interpretation of each algorithm's output.

3.2 Distributed Graph Processing Pipeline

The computational pipeline transforms raw transfer records into algorithm-ready graph structures and then into visualization-ready ranking outputs. Conceptually, the pipeline is divided into five stages:

1. **Data ingestion:** loading the transfer dataset from CSV format.
2. **Preprocessing and normalization:** parsing rows, cleaning club and league names, parsing transfer fees, and handling missing or invalid fee values.
3. **Graph construction:** aggregating transfer rows into weighted directed edges.
4. **Algorithm execution:** computing PageRank or HITS scores over the filtered graph.
5. **Post-processing and delivery:** extracting top-ranked clubs, formatting metadata, and exposing results through an API for visualization.

The implementation uses Apache Spark through the PySpark API. Spark's Resilient Distributed Dataset (RDD) abstraction is employed for the core ranking computations because both PageRank and HITS require repeated iterative propagation over graph edges. The in-memory execution model of Spark reduces the overhead associated with repeated re-computation and is therefore appropriate for iterative graph analytics.

3.2.1 Data Ingestion and Raw Representation

The original dataset is stored as a CSV file in which each row represents a player transfer event. The primary fields used in the present work are:

- player name,
- selling club,

- selling league,
- buying club,
- buying league,
- season,
- transfer fee.

The raw file is read once and then cached in memory to reduce repeated I/O overhead. Since the application may serve multiple user requests with different filters, caching the parsed dataset is an important design choice that improves responsiveness and preserves a consistent computation base across runs.

3.2.2 Preprocessing and Fee Handling

Preprocessing is necessary because real-world transfer datasets are not perfectly uniform. Club names require whitespace normalization, fee values may be missing or malformed, and repeated transfers between the same pair of clubs must be consolidated.

In the implemented system, each CSV line is parsed into structured fields. Transfer fees are converted into floating-point values. If a fee is unavailable or cannot be parsed, a fallback nominal value is assigned. This decision ensures that free transfers or incomplete records do not disappear entirely from the graph representation, while still preserving the distinction between expensive and low-value transfers.

From a methodological perspective, this preprocessing step serves two purposes:

1. it produces a graph representation compatible with iterative link-analysis algorithms;
2. it preserves as much of the empirical transfer structure as possible without overfitting the implementation to perfectly clean data.

3.2.3 Filtering Strategy

The application supports multiple filters so that the same dataset can be analyzed from different perspectives. The filtering stage is executed before the ranking algorithms are applied. The active filters are:

- **League filter:** limits analysis to transfers where both clubs belong to the selected league, thereby producing an intra-league subgraph.
- **Season range filter:** limits analysis to transfers whose season lies within an inclusive start and end boundary.
- **Weight mode filter:** selects either fee-weighted edges or count-weighted edges.

The result of filtering is a subgraph

$$G' = (V', E', W')$$

that becomes the input to either PageRank or HITS. This design ensures that both algorithms operate on exactly the same graph under the same user-selected conditions, which is essential for fair comparison.

3.2.4 Graph Construction

After filtering, individual transfer rows are aggregated into a directed weighted edge list. If multiple transfers occur from club u to club v, they are merged into a single edge (u, v) whose weight is either:

$$W(u, v) = \sum_{k=1}^{m} f_k$$

in fee-weighted mode, where f_k is the fee of the k-th transfer, or

$$W(u, v) = m$$

in count-weighted mode, where m is the number of transfers between the two clubs.

This aggregation step converts the original transfer event table into a club-level interaction graph. Methodologically, this is important because the thesis is not ranking individual players or transfer events; rather, it is ranking clubs according to their structural position within the transfer ecosystem.

3.3 Experimental Design

The comparative study is designed to isolate the effects of algorithmic formulation, edge weighting, and graph filtering. Rather than evaluating only a single ranking output, the experiments examine the stability and interpretability of rankings across multiple conditions.

3.3.1 Comparison Dimensions

The main dimensions of comparison are:

1. **Algorithm type:** PageRank versus HITS.
2. **Ranking target:** buyer-oriented versus seller-oriented analysis.
3. **Weighting scheme:** transfer fee versus transfer count.
4. **League scope:** all leagues versus selected intra-league subgraphs.

5. **Temporal scope:** full dataset versus selected season ranges.
6. **Iteration settings:** the number of iterative updates used by each algorithm.
7. **Damping factor:** applicable to PageRank only.

3.3.2 Interpretation of Buyer and Seller Rankings

Because transfer edges are directed from seller to buyer, the two algorithms are interpreted as follows:

- **PageRank (buyers):** identifies clubs that receive transfers from structurally significant sellers. These clubs function as attractors in the market.
- **PageRank (sellers):** is computed by reversing the edge direction before execution, thereby identifying clubs that supply players to structurally important buyers.
- **HITS authorities:** correspond to buyer clubs that receive links from strong hubs.
- **HITS hubs:** correspond to seller clubs that transfer players toward strong authorities.

This mapping is one of the key methodological contributions of the thesis because it adapts classical web-link semantics to a sports-economic graph without changing the underlying mathematics of the algorithms.

3.3.3 Top-N Focus

Although both algorithms can compute scores for all clubs in the graph, the thesis emphasizes the top-N ranked clubs in each run. This design decision serves both analytical and practical goals. Analytically, the top-ranked clubs are the most meaningful for comparative interpretation. Practically, visualizing the full graph would reduce readability and hinder interaction in the web interface.

3.3.4 Historical Analysis

In addition to static rankings, the methodology includes historical analysis by computing rankings season-by-season. This enables the observation of temporal changes in club prominence and reveals how major clubs rise, fall, or stabilize across different years. The historical mode therefore complements the static comparison by adding a dynamic perspective to the study.

3.4 Implementation

The implementation of this thesis consists of a full-stack analytical system named *Transfer Analyzer*. The system combines a PySpark-based ranking engine, a FastAPI backend, and a browser-based frontend for interactive visualization.

3.4.1 Technology Stack

The implemented system uses the following technologies:

- **Python / PySpark:** for data loading, preprocessing, and iterative graph algorithm execution;
- **Apache Spark:** for distributed data processing and in-memory iterative computation;
- **FastAPI:** for exposing computation endpoints to the frontend;
- **HTML, CSS, and JavaScript:** for building the interactive user interface;
- **Cytoscape.js, Chart.js, and D3.js:** for network, bar-chart, and animated historical visualizations.

3.4.2 System Architecture

Architecturally, the system follows a layered design:

1. the **data layer** stores the raw transfer CSV file;
2. the **processing layer** constructs graph structures and computes rankings using Spark RDD transformations;
3. the **service layer** exposes REST-style API endpoints;
4. the **presentation layer** renders charts, network graphs, comparative panels, and historical animations.

This separation of concerns improves maintainability and supports experimentation because algorithmic changes can be introduced in the processing layer without restructuring the interface layer.

3.5 Transfer Analyzer Web Application

The *Transfer Analyzer* serves as the user-facing artifact of the thesis. It is not merely a visualization add-on, but an essential part of the methodology because it makes the comparison operational, reproducible, and inspectable by an end user.

3.5.1 Backend Processing Workflow

When the application starts, a Spark session is initialized and the transfer dataset is preloaded into memory. For each incoming analysis request, the backend:

1. reads the selected filters and algorithm parameters;
2. builds the filtered transfer graph;
3. executes the requested algorithm;
4. extracts the top-ranked clubs and relevant edges;
5. returns a structured JSON response containing node data, edge data, and metadata.

Additional backend endpoints support league retrieval, transfer-detail lookup for specific clubs, historical ranking generation, and league-to-league flow computation.

3.5.2 Frontend Analytical Views

The frontend provides multiple complementary views of the same analytical output:

- **Bar chart view:** emphasizes ranking order and score magnitude.
- **Network graph view:** emphasizes structural relationships among top-ranked clubs.
- **Flow view:** aggregates transfers at the league level to reveal broader movement patterns.
- **Compare mode:** renders PageRank and HITS side by side under identical filter settings.
- **Historical race chart:** animates the top-ranked clubs over time.

The coexistence of these views is methodologically useful because no single visualization fully captures all aspects of graph behavior. Ranked bars show magnitude clearly, network layouts show relations, and animations expose temporal change.

3.6 Configurable Parameters

To ensure that the study is exploratory as well as comparative, the system exposes several parameters to the user:

- **Algorithm:** PageRank or HITS;
- **Target direction:** buyers or sellers;
- **Iterations:** the number of update steps performed;
- **Damping factor:** applicable to PageRank;

- **Top-*N*:** the number of clubs retained for visualization;
- **League filter:** limits analysis to a specific intra-league context;
- **Season range:** limits analysis to a selected temporal interval;
- **Weight mode:** fee-based or count-based edge weights.

This parameterization is essential for the thesis because it enables controlled comparisons. For example, changing only the weighting mode while keeping all other settings fixed provides insight into whether a club's prominence is driven by high-value transfers or by frequent transfer activity.

3.7 Algorithm Integration

Both PageRank and HITS are implemented within the same processing framework and operate on the same aggregated edge representation.

3.7.1 PageRank Integration

PageRank is implemented as an iterative weighted propagation process over outgoing edges. For a node u with outgoing neighbors v, the current rank of u is distributed proportionally according to the normalized edge weights. The update rule used in the implementation is:

$$PR_{t+1}(v) = (1-d) + d \sum_{u \in B(v)} PR_t(u) \cdot \frac{W(u,v)}{\sum_{x \in F(u)} W(u,x)}.$$

This formulation preserves the spirit of weighted PageRank while adapting it to the football transfer network. A high rank is therefore assigned to clubs that receive transfers from already influential sellers.

3.7.2 HITS Integration

HITS is implemented on the same directed graph without reversing the edge direction. Starting from uniform initialization, the algorithm alternates between:

1. updating authority scores from incoming hub contributions;
2. normalizing authority scores;
3. updating hub scores from outgoing authority contributions;
4. normalizing hub scores.

The update rules are:

$$a(v) = \sum_{u:(u,v)\in E} h(u)\, W(u,v)$$

and

$$h(u) = \sum_{v:(u,v)\in E} a(v)\, W(u,v).$$

The resulting authority and hub vectors reveal structurally different roles that are not captured by PageRank's single scalar score.

3.7.3 Unified Comparison Logic

The unified design of the system ensures that:

- both algorithms use the same filtered dataset;
- both algorithms use the same graph construction process;
- both algorithms are exposed through the same service interface;
- both algorithms can be visualized individually or side by side.

This integration is fundamental to the thesis contribution because it transforms the comparison from a conceptual discussion into a reproducible computational experiment.

3.8 Expected Analytical Outputs

The methodology is designed to produce several forms of output, each serving a distinct analytical purpose:

1. **Club rankings:** top-N lists of clubs ordered by PageRank, authority score, or hub score.
2. **Score-aware network graphs:** club interaction networks annotated by computed score values.
3. **Transfer detail panels:** club-specific summaries of important incoming and outgoing transfers.
4. **League flow summaries:** aggregated movement between leagues.
5. **Historical ranking timelines:** season-wise ranking sequences that reveal temporal evolution.
6. **Comparative insights:** agreement and divergence between PageRank and HITS under matched configurations.

These outputs are subsequently interpreted in the Results chapter to answer the research questions stated in Chapter 1. In particular, they enable the thesis to distinguish between globally central clubs, authority-like destination clubs, and hub-like supplier clubs, while also showing how these roles change across time, leagues, and weighting schemes.

Chapter 4

Results

This chapter presents the results of applying PageRank and HITS to the football transfer network and interpreting the outcomes through the *Transfer Analyzer* system. The purpose of the chapter is not merely to display rankings, but to explain what kinds of structural insights emerge when the same graph is analyzed through different algorithmic perspectives, weighting schemes, temporal filters, and visualization modes.

The results are organized around the experimental setting, the evaluation dimensions, and the major analytical views provided by the implemented web application. Since the application supports interactive exploration, the chapter also discusses how different visual modes contribute to the interpretation of ranking behavior.

4.1 Experimental Setting and Dataset

The experiments were conducted on the football transfer dataset described in Chapter 2. The dataset contains player transfer records spanning multiple seasons and leagues, with each row representing one transfer event between a selling club and a buying club. After preprocessing and aggregation, the dataset is transformed into a weighted directed graph where clubs are nodes and transfer relations are edges.

The implemented analytical system supports both full-network analysis and filtered analysis. Accordingly, the experiments in this thesis were carried out under the following settings:

- **Algorithm selection:** PageRank and HITS;
- **Direction of analysis:** buyer-oriented and seller-oriented;
- **Weighting mode:** transfer fee and transfer count;
- **League scope:** all available leagues or a selected intra-league subset;

- **Temporal scope:** all seasons or an inclusive season range;
- **Visualization scope:** top-N ranked clubs, with N configurable by the user.

The default configuration of the application emphasizes a top-ranked subset of clubs rather than the entire graph. This improves interpretability, especially in the interactive network and chart views, while still preserving the most analytically relevant outcomes of each algorithm.

4.2 Tools Used

The experiments and outputs described in this chapter were produced using the *Transfer Analyzer* system implemented in this thesis. The main tools used are:

- **PySpark and Apache Spark** for data loading, graph construction, and iterative score computation;
- **FastAPI** for exposing the ranking computations and dataset queries as backend services;
- **Chart.js** for ranked bar-chart visualizations;
- **Cytoscape.js** for interactive network graph visualizations;
- **D3.js** for the animated historical ranking view and flow-based visual exploration.

The combined use of these tools allows the thesis to move beyond static ranking tables and instead support an interactive style of experimental interpretation.

4.3 Evaluation Dimensions

The comparative interpretation of results in this thesis is based on several complementary dimensions:

1. the clubs ranked highest by each algorithm;
2. the differences between buyer-oriented and seller-oriented analysis;
3. the differences between fee-weighted and count-weighted graphs;
4. the effect of applying league and season filters;
5. the degree of overlap between PageRank and HITS rankings;
6. the visual patterns revealed through network, flow, and historical views.

These dimensions make it possible to compare not only the numerical outputs of PageRank and HITS, but also the different structural stories they tell about the football transfer market.

4.4 PageRank and HITS Ranking Behavior

4.4.1 Buyer-Oriented Analysis

When the graph is analyzed from the buyer perspective, both algorithms identify clubs that act as major attractors in the transfer market. In PageRank, a buyer club receives a high score when it attracts transfers from clubs that are themselves structurally significant in the network. This produces a notion of global importance: a club is important not simply because it buys many players, but because it buys from influential sellers.

HITS authority scores provide a related but distinct interpretation. A club becomes a strong authority when it receives transfers from strong hubs, that is, from clubs that are themselves strongly connected as suppliers. In practice, this means that authority rankings often emphasize structurally recognized destination clubs, but they may order them differently than PageRank because HITS separates the notions of receiving importance and distributing importance.

From an analytical standpoint, this distinction is significant. PageRank answers the question of which buyer clubs are globally central in the transfer ecosystem, whereas HITS authority analysis answers the question of which clubs serve as recognized destinations within a hub–authority structure.

Rank	Club	Score
1	Chelsea	11.5013
2	Man City	11.3209
3	Real Madrid	10.6357
4	Man Utd	9.7196
5	FC Barcelona	9.0998
6	Liverpool	8.4989
7	Paris SG	8.3230
8	Juventus	7.5093
9	Spurs	6.3105
10	Everton	5.8593

Table 4.1: Top 10 Buyers PageRank - Fee Weighted

Rank	Club	Score
1	FC Barcelona	0.3671
2	Man City	0.3594
3	Real Madrid	0.3364
4	Chelsea	0.3352
5	Paris SG	0.3093
6	Man Utd	0.2817
7	Juventus	0.2565
8	Liverpool	0.2165
9	Inter	0.1847
10	Atlético Madrid	0.1708

Table 4.2: Top 10 Authority Buyers HITS - Fee Weighted

4.4.2 Seller-Oriented Analysis

Seller-oriented analysis reveals another important difference between the two algorithms. In the implemented system, PageRank seller rankings are computed by reversing the edge direction so that clubs transferring players to important buyers receive higher supplier-oriented scores. This view highlights clubs that act as influential sources in the market.

HITS hub rankings naturally provide a structurally similar seller perspective without requiring a separate semantic reinterpretation of the algorithm. A club becomes a strong hub when it transfers players to strong authorities. This makes HITS particularly suitable for identifying supplier clubs that occupy an important intermediary or feeder role in the network.

The comparison between seller-oriented PageRank and hub-oriented HITS is therefore one of the most meaningful aspects of the thesis. It demonstrates that even when the two methods are aimed at a similar conceptual role, they reach that role through different mathematical mechanisms.

Rank	Club	Score
1	Inter	2.2528
2	FC Porto	2.1721
3	Parma	1.9873
4	Juventus	1.8130
5	AS Roma	1.7611
6	Atlético Madrid	1.5984
7	Liverpool	1.5394
8	Udinese Calcio	1.5087
9	São Paulo	1.4099
10	Real Madrid	1.3974

Table 4.3: Top 10 Sellers PageRank - Fee Weighted

Rank	Club	Score
1	Monaco	0.3257
2	Real Madrid	0.2551
3	FC Barcelona	0.2540
4	Liverpool	0.2431
5	Benfica	0.2119
6	FC Porto	0.2082
7	Valencia CF	0.1994
8	Arsenal	0.1986
9	AS Roma	0.1922
10	Juventus	0.1843

Table 4.4: Top 10 Hubs Sellers HITS - Fee Weighted

4.5 Effect of Weighting Mode

One of the most informative comparisons in the thesis concerns the choice of edge weighting. The same club may appear highly influential in one weighting mode and less influential in the other.

4.5.1 Fee-Weighted Analysis

In fee-weighted mode, edges represent the total monetary value transferred between clubs. This favors clubs involved in financially large transfers and therefore tends to emphasize market power, financial reach, and participation in high-value player exchanges.

Under this mode, clubs associated with elite transfer markets are expected to dominate the top ranks, since the algorithmic importance flowing through the graph is proportional to the magnitude of transfer fees. As a result, fee-weighted analysis is especially useful for studying economic centrality and transfer-market prestige.

4.5.2 Count-Weighted Analysis

In count-weighted mode, every transfer contributes equally regardless of financial size. This shifts the interpretation from economic magnitude to transactional activity. Clubs with many repeated exchanges may rise in importance even if the average value of those exchanges is relatively modest.

This distinction is valuable because it prevents the study from reducing transfer importance to money alone. Count-based analysis captures activity intensity, while fee-based analysis captures financial weight. Comparing the two reveals whether a club's prominence comes from expensive transfers, frequent transfers, or a mixture of both.

Rank	**Club**	**Score**
1	Man City	6.6434
2	Chelsea	6.3646
3	Spurs	6.2872
4	Liverpool	5.7871
5	Everton	5.2074
6	Man Utd	4.7530
7	Juventus	4.6395
8	Inter	4.4033
9	Aston Villa	4.2845
10	FC Barcelona	4.1676

Table 4.5: Top 10 Buyers PageRank - Count Weighted

Rank	**Club**	**Score**
1	Inter	0.2774
2	Juventus	0.2762
3	Chelsea	0.2511
4	Man City	0.2430
5	AC Milan	0.2210
6	FC Barcelona	0.2183
7	AS Roma	0.2178
8	Real Madrid	0.2004
9	Liverpool	0.1984
10	Spurs	0.1927

Table 4.6: Top 10 Authority Buyers HITS - Count Weighted

Rank	**Club**	**Score**
1	Inter	1.8472
2	Parma	1.5272
3	Liverpool	1.3886
4	FC Porto	1.3215
5	Udinese Calcio	1.2827
6	Juventus	1.2780
7	AS Roma	1.1992
8	NK Zagreb	1.1879
9	Feyenoord	1.1712
10	Dinamo Zagreb	1.1674

Table 4.7: Top 10 Sellers PageRank - Count Weighted

Rank	Club	Score
1	Inter	0.2200
2	Real Madrid	0.2001
3	Genoa	0.1933
4	Parma	0.1796
5	FC Porto	0.1795
6	Udinese Calcio	0.1794
7	Monaco	0.1743
8	Chelsea	0.1737
9	Juventus	0.1724
10	AS Roma	0.1713

Table 4.8: Top 10 Hubs Sellers HITS - Count Weighted

The count-weighted results strongly contrast with the fee-weighted results, particularly for HITS analysis. Under a count-based interpretation, Italian clubs like Inter, Juventus, Parma, and AS Roma emerge significantly more prominent as both authorities and hubs. This indicates a high frequency of transfer interactions within the Italian market and across Europe, regardless of the individual transfer fees involved. The hub/authority duality confirms that these clubs maintain central structural roles precisely because they acquire and supply players repeatedly, serving as key operational nodes in the talent exchange network.

4.6 Effect of League and Season Filters

4.6.1 League-Specific Analysis

Applying league filters changes the graph topology substantially. Instead of analyzing the global transfer ecosystem, the user can isolate an intra-league market and observe how centrality behaves within a narrower competitive environment.

This is methodologically important because a club's global importance does not necessarily imply equivalent importance within a specific league-only context. A club may be highly central in the broader European transfer network while appearing less dominant when only domestic league interactions are considered. Conversely, clubs that play an active role within a league may become more visible once global giants are excluded.

4.6.2 Season-Range Analysis

Season filters provide a temporal lens through which ranking behavior can be examined. Restricting the graph to specific years changes both the set of active edges and the balance of influence among clubs. This makes it possible to study how transfer-market centrality evolves over time.

Temporal filtering is particularly important for football, where club influence is not static. Management changes, financial conditions, sporting success, and regulatory shifts can all alter a club's transfer behavior. The season-range feature therefore transforms the study from a purely static graph analysis into a temporally aware investigation of market dynamics.

4.7 Visualization-Based Interpretation

The results of the thesis are not interpreted solely through raw scores. The *Transfer Analyzer* provides several analytical views, each contributing differently to the understanding of PageRank and HITS.

4.7.1 Bar Chart View

The ranked bar chart is the clearest view for comparing score magnitudes and ordinal positions. It allows immediate identification of the top clubs under each algorithm and is particularly useful for observing how the top-N ordering changes when filters or parameters are modified.

For the thesis, the bar chart serves as the most direct presentation of ranking outcomes. It is especially effective when discussing which clubs dominate under fee-weighted versus count-weighted analysis, or when comparing buyer and seller roles.

Figure 4.1: Interactive Bar Chart View displaying top-ranked clubs.

4.7.2 Network Graph View

The network view complements the bar chart by showing the actual structural relations among clubs. While a ranked list indicates which clubs score highly, the network graph reveals how those clubs are connected through transfer edges.

This view is valuable because centrality scores gain more meaning when interpreted alongside the transfer relations that produced them. A club may rank highly due to receiving transfers from many peripheral clubs, from a few highly important clubs, or through dense interaction with other top-ranked clubs. The network graph helps distinguish among these possibilities visually.

Figure 4.2: Network Graph View showing transfer relations between top clubs.

4.7.3 League Flow View

The flow-based view aggregates transfers at the league level rather than the club level. This is an important analytical extension because it shifts the perspective from individual club competition to broader market movement between leagues.

In this mode, the thesis can interpret results not only in terms of club centrality but also in terms of macro-level transfer channels. It becomes possible to observe which leagues function as dominant suppliers, which act as major destinations, and how transfer value circulates across the football ecosystem.

Figure 4.3: League Flow View illustrating transfer volume between leagues.

4.7.4 Compare Mode

The side-by-side compare mode is one of the most important contributions of the implemented system. It displays PageRank and HITS under identical filter conditions, allowing the user to inspect overlap and divergence directly.

This view supports the core research aim of the thesis: not simply to compute two algorithms independently, but to compare how they rank the same clubs on the same graph. Similar rankings suggest structural agreement, while different rankings indicate that the algorithms are emphasizing different notions of importance.

Figure 4.4: Compare Mode displaying PageRank and HITS results side-by-side.

4.7.5 Historical Animated Bar Chart

The historical animated bar chart extends the analysis across time by showing how club rankings evolve season by season. Rather than presenting isolated static snapshots, this mode visualizes the dynamic rise, decline, and persistence of clubs in the transfer market.

This temporal mode is especially useful for identifying long-term patterns. Some clubs may remain consistently prominent across many seasons, indicating sustained market influence. Others may experience short-lived surges corresponding to specific financial periods, sporting projects, or transfer strategies. The animated view therefore transforms the results into a narrative of historical evolution rather than a single final ranking.

Figure 4.5: Historical Animated Bar Chart tracking rankings across seasons.

4.7.6 Case Study: FC Barcelona Across the Years

To move beyond aggregate comparison and demonstrate the practical value of the developed system, this thesis also considers a concrete club-level case study. FC Barcelona is an appropriate example because it is one of the most visible clubs in the dataset and because its role in the transfer market changes substantially across years, making it well suited for longitudinal analysis.

Using the historical ranking logic of the implemented system, FC Barcelona was tracked season by season under the same algorithmic settings. The resulting pattern shows that the club is not uniformly dominant across the whole timeline; rather, its prominence rises strongly in some periods and declines sharply in others depending on the algorithm and weighting mode.

Under **fee-weighted PageRank**, FC Barcelona appeared in the top 10 in 8 out of the 19 analyzed seasons and entered the top 5 in 3 seasons. Its strongest period occurred in the late 2000s and mid-2010s. In particular, FC Barcelona ranked **2nd** in 2008 and 2009 and reached **1st place** in 2014. By contrast, it fell far outside the top ranks in several other seasons, including 2002, 2013, 2015, and 2018. This behavior indicates that FC Barcelona's global transfer-market centrality was highly concentrated in certain high-impact periods rather than being uniformly dominant throughout the entire dataset.

Under **fee-weighted HITS authority**, the picture is different. FC Barcelona appeared in the top 10 in 9 seasons and in the top 5 in 7 seasons, which is more stable than its PageRank top-5 presence. It ranked **2nd** in 2000, 2004, 2010, 2011, and 2017, and remained highly visible as an authority club in several other years even when its PageRank position was lower. This suggests that FC Barcelona repeatedly functioned as a structurally important *destination* club, receiving players from well-connected supplier clubs, even in seasons where it was not globally dominant in PageRank terms.

The contrast becomes even clearer when count-based weighting is considered. Under **count-**

weighted PageRank, FC Barcelona reached the top 10 in only 4 seasons and the top 5 in only 2 seasons, with its best result again being **1st** in 2014. Under **count-weighted HITS authority**, however, it still appeared in the top 10 in 8 seasons, including **2nd** place in 2000 and 2009 and strong positions in 2004 and 2006. This indicates that FC Barcelona's structural role is more consistently captured by authority-based analysis than by transaction-frequency-based global centrality.

Taken together, these results support an important substantive conclusion. FC Barcelona's importance in the transfer network is not explained simply by the number of transfers it performs. Instead, its prominence is more strongly associated with the *quality and structural importance* of the clubs from which it acquires players and, in its strongest years, with its participation in financially significant transfer exchanges. In other words, FC Barcelona is better characterized as a repeatedly strong **authority** club and a periodically dominant **PageRank** club, rather than as a consistently high-volume transfer actor.

Algorithm & Weight	Total Top-10	Total Top-5	Best Rank	Best Year(s)
PageRank (Fee)	8	3	1st	2014
HITS Authority (Fee)	9	7	2nd	2000, 2004, 2010, 2011, 2017
PageRank (Count)	4	2	1st	2014
HITS Authority (Count)	8	4	2nd	2000, 2009

Table 4.9: Summary of FC Barcelona's Rankings Across Seasons (Out of 19 Total Seasons)

This case study illustrates the broader value of the thesis approach. If only one algorithm had been used, the interpretation of FC Barcelona's historical role would have been incomplete. PageRank highlights the years in which the club occupied a globally central position in the weighted transfer ecosystem, whereas HITS authority highlights the years in which it acted as a major destination for talent from structurally important suppliers. The season-by-season analysis therefore reinforces the thesis's central argument that transfer-market importance is dynamic, multi-dimensional, and dependent on the ranking method used.

4.8 Comparative Discussion of Findings

The most important result of the thesis is that PageRank and HITS do not simply provide duplicate rankings under different names. Instead, they illuminate different structural aspects of the same transfer network.

PageRank is more naturally interpreted as a measure of global transfer-market centrality. It identifies clubs that occupy influential positions in the overall directed flow of player movement. HITS, by contrast, provides a two-role decomposition that distinguishes destination power from supplier power. This distinction is particularly useful in football transfer analysis because the market contains clubs that mainly attract talent, clubs that mainly supply talent, and clubs that participate in both roles.

The weighting comparison further strengthens this conclusion. Fee-based analysis emphasizes economic influence, while count-based analysis emphasizes interaction frequency. League and season filters then show that these notions of influence are context dependent rather than universal. Together, these findings support the central claim of the thesis: the value of comparing PageRank and HITS lies not in choosing a single winner, but in understanding how each algorithm highlights a different dimension of network structure.

4.9 Chapter Summary

This chapter presented the results framework for interpreting PageRank and HITS on the football transfer network. The analysis considered buyer and seller perspectives, fee-based and count-based weighting, league and season filtering, and multiple visualization modes including ranked charts, network graphs, league flows, compare mode, and historical animation.

Taken together, the results show that the football transfer market is not adequately described by a single notion of importance. Instead, different algorithmic and visual perspectives reveal different forms of influence, centrality, supply behavior, and temporal evolution. The next chapter concludes the thesis by summarizing the main contributions and outlining directions for future work.

Chapter 5

Conclusion

5.1 Summary of the Thesis

This thesis presented a comparative study of the PageRank and HITS algorithms within the context of distributed graph processing using Apache Spark. While both algorithms were originally developed for hyperlink analysis and web search, the work demonstrated that their mathematical foundations can be meaningfully transferred to a non-web domain, namely the professional football transfer market. By modeling clubs as nodes and transfer relations as weighted directed edges, the thesis transformed football transfer activity into a graph structure suitable for large-scale link analysis.

The study addressed both theoretical and practical dimensions. On the theoretical side, it reviewed the foundations of PageRank and HITS and clarified the distinction between PageRank's global notion of importance and HITS's dual hub–authority interpretation. On the practical side, it developed a complete analytical framework that allows both algorithms to be executed on the same dataset under shared filtering and weighting conditions, thereby enabling direct comparison.

5.2 Main Contributions

The main contributions of the thesis can be summarized as follows:

1. **A unified comparative framework.** The thesis established a common analytical environment in which PageRank and HITS can be applied to the same weighted directed graph and interpreted through the same set of filters, parameters, and visualization modes.
2. **A graph-based formulation of the football transfer market.** The work formalized football transfers as a weighted directed network, making it possible to analyze the mar-

ket using graph-ranking methods rather than relying solely on descriptive or financial summaries.

3. **A distributed implementation using PySpark.** The thesis implemented both algorithms using Apache Spark's distributed processing model, providing a scalable computational foundation for iterative graph analysis.

4. **An interactive analytical application.** The *Transfer Analyzer* web application exposed the ranking results through bar charts, network graphs, league-flow views, side-by-side comparison mode, and a historical animated ranking view.

5. **A domain-specific interpretation of graph rankings.** The thesis translated the abstract outputs of PageRank and HITS into football-specific meanings such as buyer attractors, supplier clubs, authorities, hubs, and league-level transfer dynamics.

5.3 Comparative Conclusions

The central conclusion of the thesis is that PageRank and HITS do not produce interchangeable views of the football transfer market. Instead, they illuminate different structural properties of the same network.

PageRank is most naturally interpreted as a measure of global transfer-market centrality. It highlights clubs that occupy influential positions in the broader directed flow of player movement. HITS, by contrast, provides a role-based decomposition that distinguishes destination power from supplier power through authority and hub scores. This distinction is especially valuable in football transfer analysis because the market contains clubs that mainly attract talent, clubs that mainly supply talent, and clubs that participate in both roles to different degrees.

The comparative analysis also showed that algorithmic interpretation depends strongly on the graph model chosen. Fee-weighted analysis emphasizes financial magnitude and high-value transfers, while count-weighted analysis emphasizes repeated activity and transactional frequency. League filters and season filters further demonstrate that centrality is not universal but depends on competitive context and temporal scope. Together, these observations support the broader claim that the value of comparing PageRank and HITS lies not in selecting a single superior algorithm, but in understanding how each one reveals a different dimension of network structure.

5.4 Final Remarks

Overall, the thesis showed that the football transfer market is a compelling application domain for graph-ranking methods because it combines directional exchange, financial weight, structural asymmetry, and temporal evolution within a single real-world network. By integrating theory, distributed implementation, and interactive analysis, the work contributes both a conceptual comparison of PageRank and HITS and a practical system for exploring their behavior.

The resulting framework offers a reproducible foundation for future graph-based studies of football economics and, more broadly, of other complex socioeconomic systems that can be represented as weighted directed networks.

Chapter 6

Future Work

Although the current thesis provides a strong foundation for comparing PageRank and HITS on the football transfer network, several important extensions remain possible. These directions would strengthen both the technical system and the analytical depth of the study.

6.1 Algorithmic Extensions

One natural direction for future work is to expand the comparison beyond PageRank and HITS. Other graph-based ranking and centrality methods, such as eigenvector centrality, Katz centrality, SALSA, or community-aware ranking approaches, could be integrated into the same framework. This would allow the football transfer market to be studied through a broader range of structural perspectives.

Another promising extension is the incorporation of personalized or topic-sensitive variants of PageRank. In the football context, these variants could be adapted to emphasize particular leagues, club tiers, geographical regions, or player categories, thereby producing more specialized notions of influence.

6.2 Data Enrichment

The current thesis focuses on transfer relations weighted by fee or count. Future versions of the dataset could be enriched with additional attributes such as player age, position, nationality, market value, contract duration, or club performance in the same season. This would make it possible to build more nuanced graph representations and study specific market dimensions in greater detail.

An especially valuable extension would be the integration of more recent seasons and continuous dataset updates. Doing so would transform the current historical analysis into a living analytical system capable of tracking ongoing transfer-market behavior.

6.3 Temporal and Dynamic Modeling

Although the present work includes season-based filtering and animated historical ranking views, the underlying algorithms are still applied to static snapshots of the graph. Future work could investigate genuinely dynamic graph models in which scores evolve continuously over time rather than being recomputed independently for each season.

Such an approach would support more rigorous analysis of persistence, abrupt structural shifts, and long-term club trajectories. Dynamic centrality models could help distinguish temporary transfer surges from sustained institutional influence.

6.4 Scalability and System Improvements

From a systems perspective, the current implementation is well suited for experimentation and interactive exploration, but it could be extended to support larger datasets, multi-node Spark deployment, and more advanced caching or persistence strategies. Running the application over a true Spark cluster would enable more explicit study of scalability under larger graph sizes and heavier workloads.

The frontend could also be enhanced with downloadable reports, figure export, ranking table export, and session saving. Such features would be especially useful for research, teaching, and demonstration purposes.

6.5 Analytical Validation

Another important direction for future work is the addition of stronger quantitative validation. Future studies could compare algorithmic rankings with external football indicators such as club revenues, squad valuations, transfer spending, league standings, or European competition performance.

This would provide a more formal basis for evaluating whether graph-based importance aligns with conventional indicators of sporting or financial strength. It could also clarify when the algorithms confirm expected hierarchies and when they reveal more subtle structural roles.

6.6 Generalization to Other Domains

Finally, the framework developed in this thesis is not limited to football transfer networks. The same methodology could be applied to other directed weighted systems such as employee mobility networks, citation graphs, trade networks, supply chains, and online influence networks. Extending the framework to such domains would further demonstrate the general value of combining distributed graph processing with interactive comparative visualization.

In this sense, the thesis can be viewed not only as a study of football transfers, but also as a reusable blueprint for comparing graph-ranking algorithms on real-world complex networks.

Appendix

Bibliography

[1] Sergey Brin and Lawrence Page. The anatomy of a large-scale hypertextual web search engine. In *Proceedings of the 7th International Conference on World Wide Web (WWW7)*, pages 107–117, Brisbane, Australia, 1998. Elsevier.

[2] Lawrence Page, Sergey Brin, Rajeev Motwani, and Terry Winograd. The PageRank citation ranking: Bringing order to the web. Technical Report 1999-66, Stanford InfoLab, 1999. Available at `http://ilpubs.stanford.edu:8090/422/1/1999-66.pdf`.

[3] Jon M. Kleinberg. Authoritative sources in a hyperlinked environment. *Journal of the ACM*, 46(5):604–632, 1999.

[4] Xiao-Fan Liu, Yi-Lu Liu, Xiao-He Lu, Qin-Xia Wang, and Tian-Xiang Wang. The anatomy of the global football player transfer network: Club functionalities versus network properties. *PLoS ONE*, 11(6):e0156504, 2016. Available at `https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4892674/`.

[5] Guannan Xu and Tsuyoshi Murata. Network analysis of the transfer market of european football leagues. *IEICE Transactions on Information and Systems*, E99-D(11):2745–2752, 2016.

[6] Omar A. Guerrero and Robert L. Axtell. Employment growth through labor flow networks. arXiv preprint arXiv:1310.2642, 2013. Available at `https://arxiv.org/abs/1310.2642`.

[7] Joseph E. Gonzalez, Reynold S. Xin, Ankur Dave, Daniel Crankshaw, Michael J. Franklin, and Ion Stoica. GraphX: Graph processing in a distributed dataflow framework. In *Proceedings of the 11th USENIX Symposium on Operating Systems Design and Implementation (OSDI 14)*, pages 599–613. USENIX Association, 2014. Documentation: `https://spark.apache.org/graphx/`.

[8] Islam El-Maddah. Project 12: Comparing PageRank and HITS using Apache Spark. Project brief, Cluster 3 — Web Graph Analytics & PageRank, 2024. Supervised project description. Contact: `islam_elmaddah@yahoo.co.uk`.

[9] Neelam Duhan and A.K. Sharma. A comparative analysis of PageRank and HITS web link analysis algorithms. *International Journal of Advanced Research in Computer and Communication Engineering (IJARCCE)*, 2(2):1001–1007, 2013.

[10] Sandeep Kumar and Maninder Singh. Comparative analysis of PageRank and HITS algorithms. *International Journal of Engineering Research & Technology (IJERT)*, 3(6):357–361, 2014.

[11] Priya Sharma and Rahul Gupta. HITS vs. PageRank: A comparative study in the era of modern web architectures. *International Journal of Computer Applications (IJCA)*, 186(1):12–20, 2024.

[12] László Csató and Csaba D. Tóth. PageRank on gluing networks. *Mathematics (MDPI)*, 9(15):1773, 2021.

[13] Cornell University, Department of Computer Science. Limitations of PageRank and link analysis. Course notes, Cornell University, 2015. Available at `https://blogs.cornell.edu/info2040/`.

[14] Salome Wairimu. Limitations of the PageRank algorithm. Technical blog post, 2022. Online; accessed 2024.

[15] Taher H. Haveliwala. Topic-sensitive PageRank. In *Proceedings of the 11th International Conference on World Wide Web (WWW '02)*, pages 517–526, Honolulu, Hawaii, USA, 2002. ACM. Also referenced via ResearchGate.

[16] Taher H. Haveliwala. Topic-sensitive PageRank: A context-sensitive ranking algorithm for web search. *IEEE Transactions on Knowledge and Data Engineering*, 15(4):784–796, 2003. Stanford University.

[17] Wenpu Xing and Ali Ghorbani. Weighted PageRank algorithm. In *Proceedings of the 2nd Annual Conference on Communication Networks and Services Research (CNSR)*, pages 305–314. IEEE, 2004. Also described at `https://www.geeksforgeeks.org/weighted-page-rank-algorithm/`.

[18] Text REtrieval Conference (TREC). Weighted link analysis for web search. TREC Proceedings, 2004. Available at `https://trec.nist.gov/`.

[19] Zoltán Gyöngyi, Hector Garcia-Molina, and Jan Pedersen. Combating web spam with TrustRank. In *Proceedings of the 30th International Conference on Very Large Data Bases (VLDB)*, pages 576–587, Toronto, Canada, 2004. VLDB Endowment.

[20] Grzegorz Malewicz, Matthew H. Austern, Aart J.C. Bik, James C. Dehnert, Ilan Horn, Naty Leiser, and Grzegorz Czajkowski. Pregel: A system for large-scale graph processing. In *Proceedings of the 2010 ACM SIGMOD International Conference on Management of Data*, pages 135–146. ACM, 2010.

[21] John Haltiwanger, Henry Hyatt, Lisa B. Kahn, and Erika McEntarfer. Cyclical job ladders by firm size and firm wage. Working paper, EconStor, 2018. Available at `https://www.econstor.eu/`.

[22] Jesper Bagger and Rasmus Lentz. An empirical model of wage dispersion with sorting. Working paper, Duke University, Department of Economics, 2019. Available at `https://econ.duke.edu/`.

[23] York University, Department of Electrical Engineering and Computer Science. Link analysis: PageRank and HITS. Lecture slides, 2018. Available at York University course materials.

[24] Wikipedia contributors. PageRank — Wikipedia, the free encyclopedia. `https://en.wikipedia.org/wiki/PageRank`, 2024. Online; accessed 2024.

[25] Jing Gong and Yi Li. Network analysis of football transfer market. ResearchGate, 2020. Available at `https://www.researchgate.net/`.